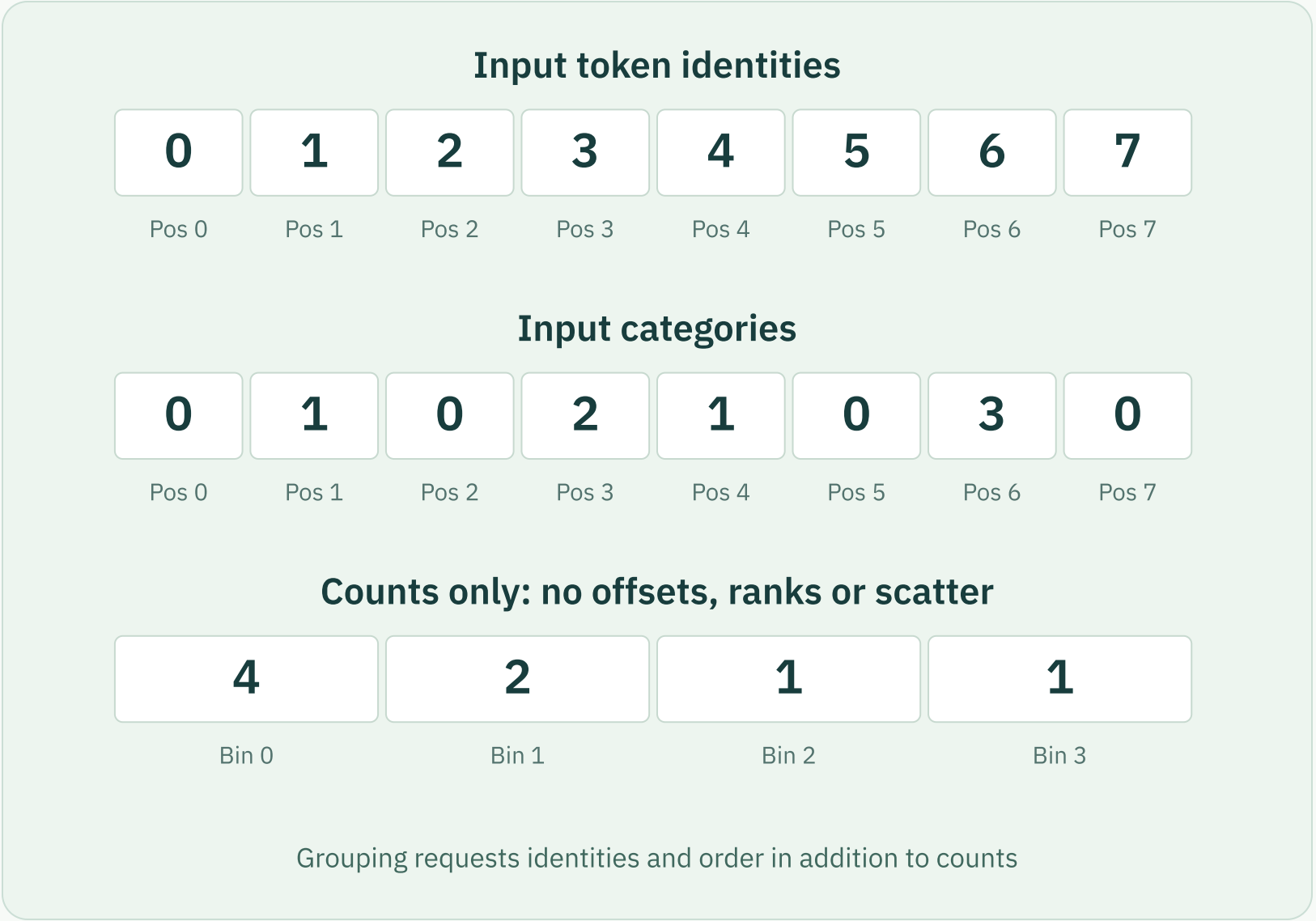
1 / 6

# Choose the required output first.

Counts describe category sizes; grouping also preserves identities.

> Local token placement and output contracts · no GPU timing or fastest algorithm





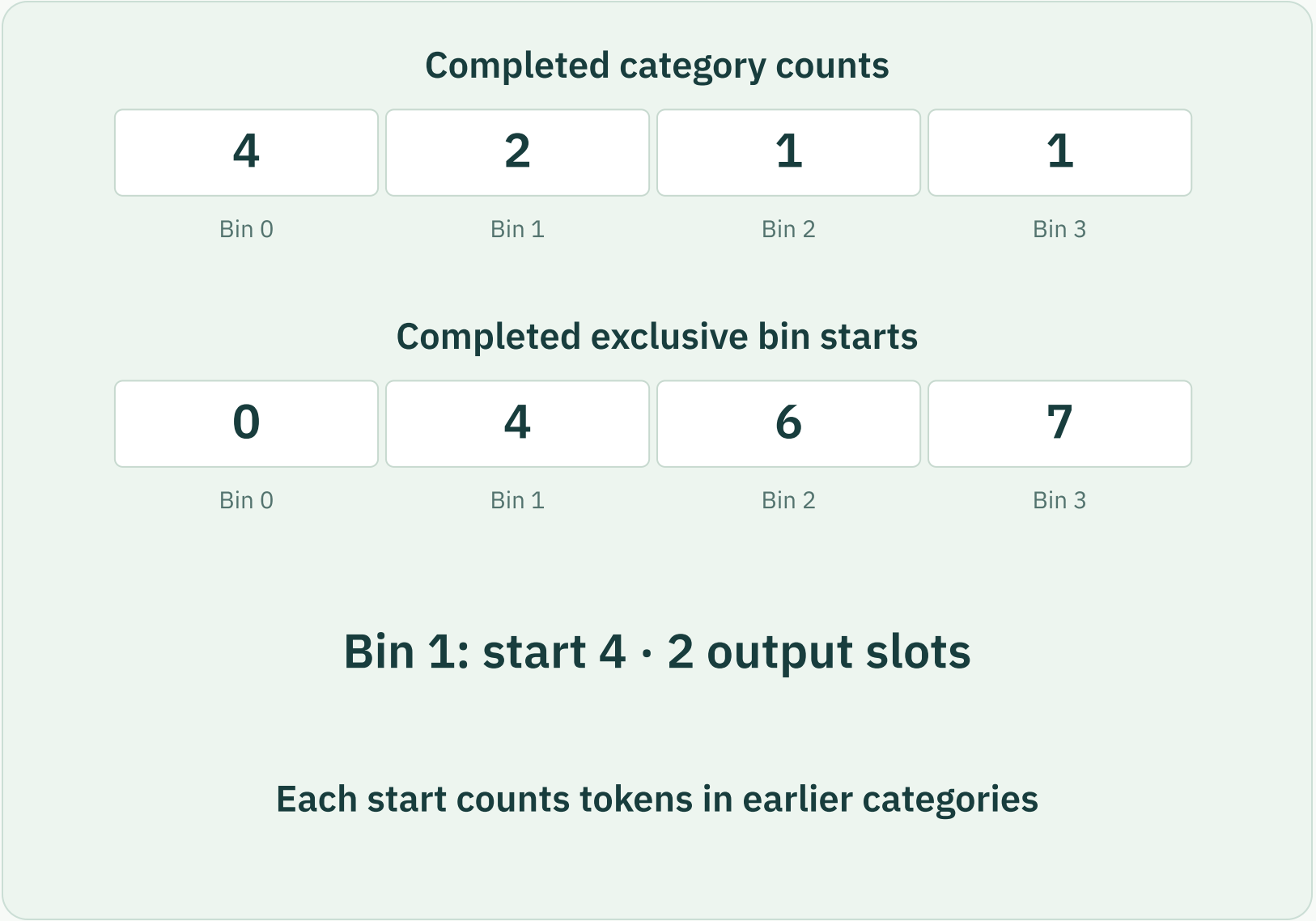

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# Counts allocate disjoint regions.

Exclusively scan completed category counts for bin starts.

> Local token placement and output contracts · no GPU timing or fastest algorithm





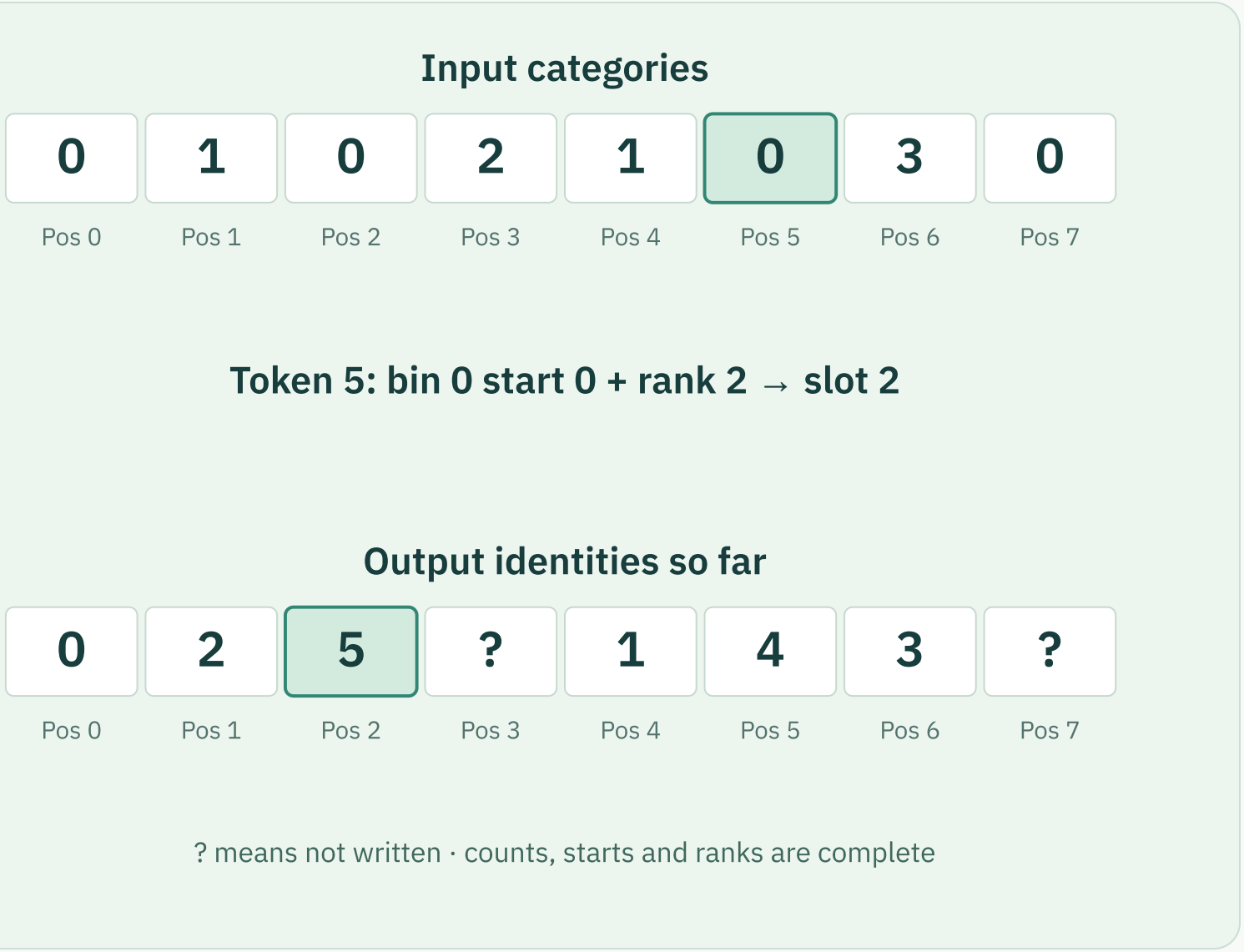

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# Add a rank to place one token.

Token 5 uses bin start 0 plus rank 2, giving output slot 2.

> Local token placement and output contracts · no GPU timing or fastest algorithm





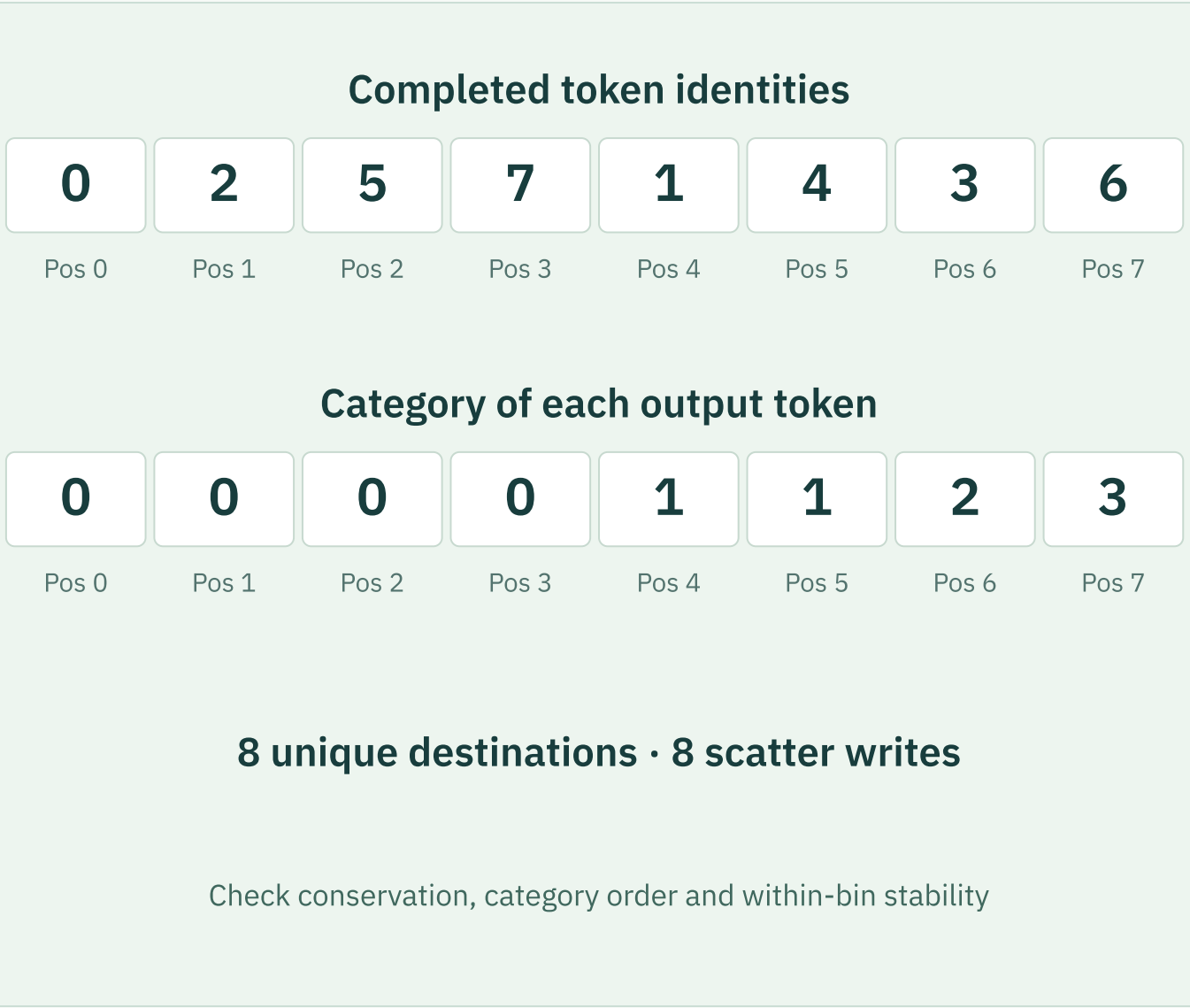

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# Check identities and stability.

Equal counts alone do not establish the required grouped result.

> Local token placement and output contracts · no GPU timing or fastest algorithm





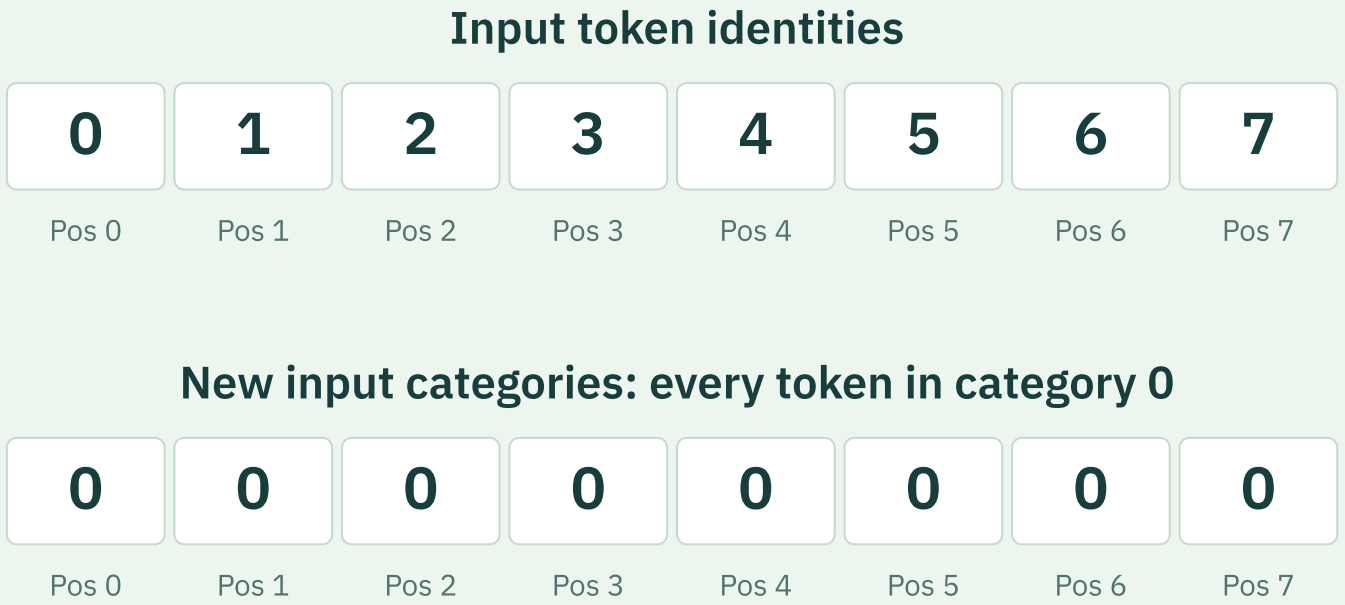

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# Pause: concentrate every token.

Keep stable grouping; change only the input categories to eight zeros.

> Local token placement and output contracts · no GPU timing or fastest algorithm





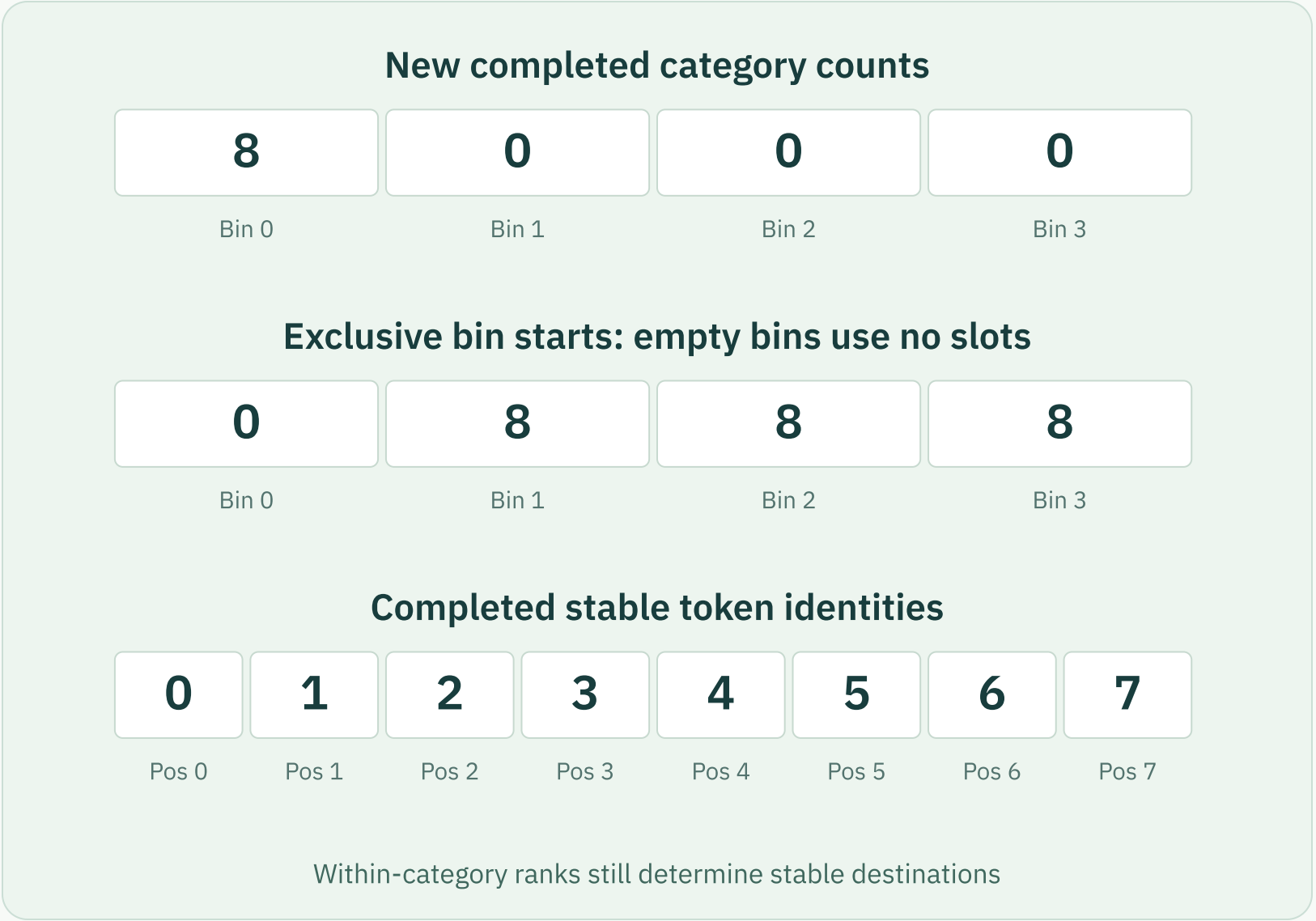

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# One occupied bin still needs ordered identities.

Empty bins start at the output end and make no writes.

> Local token placement and output contracts · no GPU timing or fastest algorithm






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